Bus Reservation And Ticketing System

https://github.com/aiyshwary/Bus-Reservation-and-Ticketing-System

Java OOP Console Application

OVERVIEW

A Java-based console application that simulates a complete bus ticket reservation and billing workflow, supporting multi-passenger booking, seat management, and discount-based billing operations.

IMPACT

Built a fully functional console-based Bus Reservation and Ticketing System in Java with structured input validation, in-memory seat management, and automated billing with passenger-type discounts.

TECHNICAL WALKTHROUGH

Java

OOP

Console Application

OVERVIEW

IMPACT

TECHNICAL WALKTHROUGH
“I built a console-based Bus Reservation and Ticketing System in Java that
allows seat management, passenger booking, billing with discounts, and transaction tracking using

- structured input validation and in-memory data

- storage. ”

Problem statement
Manual ticket booking is error-prone and slow.
This project simulates a real-world bus reservation system where users can manage destinations, seat availability, passenger bookings, billing, and payment status through a menu-driven interface.
High-level system flow

Explain this in steps

Step 1: Authentication

Simple login using username and password

Prevents unauthorized access
“This simulates role-based access at a basic level.”

Step 2: Main Menu

The system has 5 main modules

Destination Management

Passenger Booking

Billing & Payment

View Passenger Details

Exit

Module-wise explanation

Destination Module

- Displays available destinations

- Shows

Fare per destination

Remaining seat count

Seats are initialized dynamically
“Seat availability updates in real time after each booking.”

Passenger Booking Module
This is the core logic.

What happens here
Passenger name validation (no duplicates)

Destination selection with validation

Seat availability check

Passenger count input

Discount handling

Fare calculation

Business logic

    Total Fare = (Regular passengers × fare) + (Discounted passengers × (fare × 0.80))

“This ensures accurate billing while enforcing seat limits.”

Billing & Payment Module

- Search passenger by name

- Prevent double payment

- Accept payment

- Compute change

- Mark passenger as PAID

“Once paid, the passenger status is locked to avoid duplicate billing.”

View Module

- Search passenger details

- Displays

Destination

Passenger count

Discount count

Total fare

Payment status
Amount paid and change (if applicable)
“This acts like a transaction history lookup.”

Exit Module

Gracefully exits the system

Confirms continuation

KEY DETAILS

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1. Implements user authentication, seat selection, and multi-passenger booking in a single session.

2. Handles billing logic including discount tiers for different passenger categories.

3. Uses structured in-memory data storage for transaction tracking without a database dependency.

4. Validates all inputs to prevent double-booking and illegal seat selections.

5. Supports 5 Mindanao route destinations with preset fares and 20 seats each; applies a flat 20% discount for PWD, Student, and Senior Citizen passengers.

6. Offers a 5-option console menu: Destination (view routes and live seat counts), Passengers (book), Billing (payment with change), View (search and receipt), and Exit.

7. Enforces a 3-attempt login lockout; uses three parallel 2D arrays to store passenger name/destination/status, counts, and fare data.